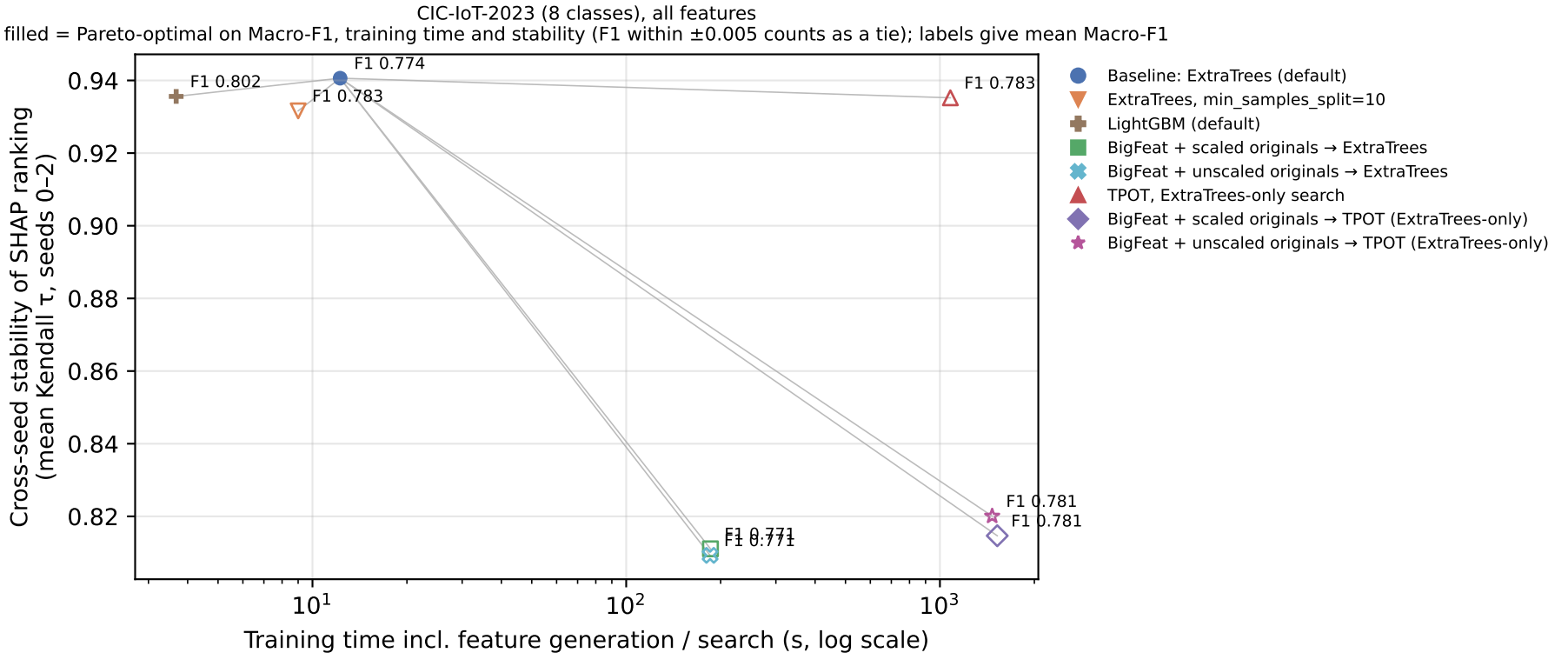
CIC-IoT-2023 (8 classes), all features
filled = Pareto-optimal on Macro-F1, training time and stability (F1 within ±0.005 counts as a tie); labels give mean Macro-F1
Cross-seed stability of SHAP ranking
(mean Kendall τ, seeds 0–2)
0.94
0.92
0.90
0.88
0.86
0.84
0.82
10^1
10^2
10^3
Training time incl. feature generation / search (s, log scale)
F1 0.802
F1 0.774
F1 0.783
F1 0.783
F1 0.781
F1 0.781
F1 0.771
F1 0.771
Baseline: ExtraTrees (default)
ExtraTrees, min_samples_split=10
LightGBM (default)
BigFeat + scaled originals → ExtraTrees
BigFeat + unscaled originals → ExtraTrees
TPOT, ExtraTrees-only search
BigFeat + scaled originals → TPOT (ExtraTrees-only)
BigFeat + unscaled originals → TPOT (ExtraTrees-only)